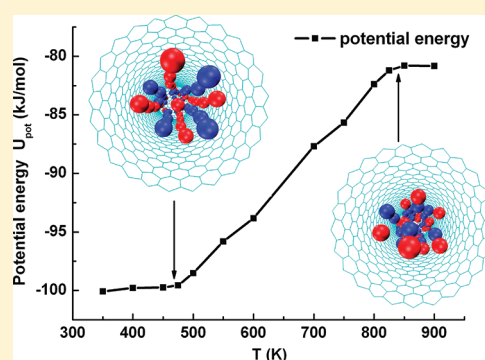


Melting Transition of Ionic Liquid [bmim][PF₆] Crystal Confined in Nanopores: A Molecular Dynamics Simulation

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Supporting Information

ABSTRACT: In this paper, the melting process of 1-butyl-3-methylimidazolium hexafluorophosphate ([bmim][PF₆]) crystals confined in carbon nanotubes were investigated using molecular dynamics simulation. The confined ions formed a “shell-chain” structure within the nanopore. Our results revealed that this “shell-chain” structure possessed long-range crystalline order at low temperature and initiated melting at approximately 500 K, which well fit with experimental observations of ionic liquids in carbon nanotubes. This melting process was also confirmed by the potential energy profile of the confined ions. It was found that below 500 K, the ions within the nanopores were almost frozen around their positions. As the temperature increased above 500 K, the average number of hydrogen bonds for each confined anion began to decline linearly. This decline led to a dramatic change in the packing arrangement of the confined ions, followed by a steep rise in the ionic diffusivity.



INTRODUCTION

Understanding the behaviors of liquids in nanoenvironments is relevant to bioscience, chemistry, geology, and materials science.^{1–4} When liquids are confined to length scales commensurate with molecular dimensions, properties significantly different from those of the bulk counterparts are commonly observed.⁵ These nanoconfinement characteristics are key elements for the development of nanoscale applications for selective ion conduction,⁶ chemical and biological sensing,^{5,7} programmable catalysis,⁸ and energy conversion.^{9,10} Additionally, interest in nanofluidics has increased with the widespread availability of carbon nanotubes, which appeared to have the ideal characteristics for these type of studies.^{7,11,12} The natural scale of nanotube channels appeared to yield the possibility of bypassing the complexities of microfabrication and obtaining tubular structures with diameters as small as 1 nm.² As a result, numerous research groups have been motivated to investigate the thermodynamics and dynamics of fluids confined in these materials. For instance, new ice phases, including ordered ice nanotubes formed in carbon nanotubes have been reported using nuclear magnetic resonance (NMR),¹³ neutron diffraction,¹⁴ vibrational spectroscopy,¹⁵ and molecular dynamics (MD) analyses.^{11,16,17} The thermodynamics of adsorption and capillary condensation of light alkanes and benzene in single-walled carbon nanotubes have also been widely studied.^{18,19} Despite the significant progress in understanding the physical properties of simple fluids, behaviors of more complex fluids confined to a nanometer scale have received considerably less attention and remain to be clarified.

In recent years, interest in ionic liquids has grown rapidly due to their desirable properties and various applications in many fields.^{20–22} Besides being extensively used in various homogeneous environments, ionic liquids also exhibit exciting application potential in multiphase and confined systems, particularly in the areas of composite materials and electrochemistry. Aida and co-workers found that as single-walled carbon nanotubes were mixed with an excess amount of imidazolium-based ionic liquids, gelatinous materials (“bucky gels”) were formed, which show good mechanical and electronic properties.²³ Then, they developed a highly elastic conductor composed from single-walled carbon nanotubes, an ionic liquid, and a compatible fluorinated copolymer, which might have important application in high-performance and large-area electronic circuits.²⁴ Recently, new composite electrodes have been fabricated by Kachooosangi et al. using multiwall carbon nanotubes and ionic liquids and shows very attractive electrochemical performances compared to other conventional electrodes, notably improved sensitivity and stability.²⁵ Room-temperature ionic liquids are also attractive electrolytes for nanoporous carbon supercapacitors, and relevant theoretical models were proposed by Huang et al.²⁶ and Kim et al.²⁷ to understand the behavior of electrolyte ions inside nanoconfined spaces.

Compared to the rapidly growing interest in composite of carbon nanotubes and ionic liquids, the behavior of ionic liquids

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confined at the nanometer scale has received scant attention. Our differential scanning calorimetry and X-ray diffraction results demonstrated that the 1-butyl-3-methylimidazolium hexafluorophosphate ([bmim][PF₆]) experienced a fully different phase transition and crystal formation when confined in carbon nanotubes.²⁸ Recently, an ordered ion arrangement of [bmim][PF₆] was observed inside the channels of carbon nanotubes using molecular dynamics simulation by Zhang et al.²⁹ and Hung et al.³⁰ A similar ordered ion distribution of 1-ethyl-3-methylimidazolium tetrafluoroborate ([emim][BF₄]) confined in carbon nanotube was also reported by Kim et al.,³¹ and cation and anion structures inside the nanotubes varied dramatically with the nanotube size. Despite this progress, ion packing, strength of cohesive interactions, and phase behavior for ionic liquids inside nanopores have not been clearly elucidated. In the current study, we employed molecular dynamics simulations to investigate the structure and melting process of [bmim][PF₆] crystals confined in a carbon nanotube over a wide temperature range. Our simulation results have revealed that inside the nanopore, cations and anions aggregate in a highly ordered manner. As temperature increases beyond 500 K, a clear and drastic solid-to-liquid transition of ionic liquid occurs.

Simulation Methodology and Model. The [bmim][PF₆] ionic liquid was treated as a systematic OPLS-AA force field as developed by Lopes et al.,³² which was successfully used to simulate the nanostructures of confined ionic liquids.^{27,30,31,33,34} For completeness, the form of the total potential energy and part of the force field parameters used in this work are given in the Supporting Information. The simulations were carried out using the Gromacs-MD package for the temperature from 350 to 950 K. We also ran several simulations at higher temperatures, and the system was stable even at $T = 1000$ K. During simulation, one close-ended carbon nanotube having a length of $L = 7.0$ nm and pore diameter $D = 1.94$ nm was immersed in an equilibrated periodic liquid reservoir containing 600 pairs of bmim⁺ cations and PF₆[−] anions. Without any filling, an initial optimization of 200 ps at the pressure of 1 atm and the temperature of 400 K was performed to remove bad contacts resulting from the initial random configuration of ionic liquids and nanotubes. After the preequilibration was finished, one cap of the close-ended carbon nanotube was removed and the ionic liquid was allowed to fill into the nanopore from the bulk outer liquid. The filling process was completed after about 10 ns, and then the system was equilibrated for 5 ns (Figure S1 in the Supporting Information). We also ran simulations by filling ionic liquids into several nanotubes of different lengths ($L = 3.0, 5.0, 10.0$ nm) and found that as the nanotube length was not shorter than 5.0 nm, computed results were similar to those obtained with $L = 7.0$ nm. At each thermodynamic point, we first conducted simulations in the NPT ensemble at 1 atm for 10 ns. After that, simulations in the canonical ensemble were carried out with 5 ns equilibration, and the data was collected for an additional 5 ns.

In all of the simulations, the bond length was constrained with the LINCS algorithm.³⁵ The nonbonded interactions were calculated using a 1.5 nm atom-based cutoff, correcting for the long-range electrostatics using the Ewald summation method (Particle Mesh Ewald “PME” approximation).³⁶ The temperature was monitored by coupling the system to a thermal bath using the Berendsen algorithm with a relaxation time of 0.1 ps.³⁷ The carbon nanotube was fixed in our simulations. The carbon atoms of carbon nanotube were also modeled as uncharged Lennard-Jones particles by the OPLS-AA force field.^{38,39} All of

the detailed parameters of the carbon atoms and ionic liquid are available in Table S1 in the Supporting Information. The parameters ϵ_{ii} and σ_{ii} for like-atom interactions were taken from the force field of Lopes et al.³² The cross term parameters for i, j interactions were obtained from the conventional combination rules $\epsilon_{ij} = (\epsilon_{ii} \cdot \epsilon_{jj})^{1/2}$ and $\sigma_{ij} = (\sigma_{ii} \cdot \sigma_{jj})^{1/2}$.

RESULTS AND DISCUSSION

Ionic Distribution and Phase Behavior in a Cylindrical Nanopore. As observed in nanofluid research, the presence of a solid wall usually induces ordering in the adjacent liquid, which is manifested in density oscillations that extend several molecular diameters.⁴⁰ In this simulation, confinement effects on the ionic liquid structure in nanopore can also be emphasized by the radial number density profiles of the confined atoms computed along the pore radius in Figure 1. The sharp peaks of the density profiles in Figure 1a,b indicate that at low temperature ions self-organize into two concentric layers in the nanopore with cations and anions composing the “shell” layer in nearly equal amounts and the interior quasi-one-dimensional chain being formed by cations only. Such surface-induced layering has already been observed in various systems with different types of interactions from simple Lennard-Jones to hydrogen-bonded systems.^{2,18,41} The effect of temperature on this “shell-chain” structure also manifests itself in changes of the radial number density profiles. As the peaks on the profiles in Figure 1c indicate, below 500 K, the tubular structure of confined ionic liquid is weakly affected by temperature. However, as the temperature increases, continuous changes may be observed. The peaks related to “shell” ions near the nanopore wall become broader and less intense. These changes could be understood by taking into account that the ions are more mobile and the ordered packing may suffer serious damage at higher temperatures. In addition, Figure 1d shows the maxima intensity of the profiles for “shell” ionic liquid as a function of temperature, suggesting that the confined ion cluster nanostructure may undergo a continuous change above 500 K.

Nanometer scale confinement may also affect the ion distribution in the axial direction as well as in the radial direction. To probe whether ordering is present beyond the radial direction, the interionic spacing distribution function is introduced in this paper. It is defined as the distance distributions between the mass centers of confined cations and anions. Considering the complicated structure and finite number of confined cations and anions, the arrangement of ions in the nanotube can be determined by the interionic spacing distribution function more directly than the radial distribution function, which is generally useful for analyzing the aggregation structure of matter. Figure 2 exhibits the interionic spacing distribution function at different temperatures. A large peak at 0.5 nm that corresponds to the first nearest neighbor shell formed by counterions is observed. According to the definition of the interionic spacing distribution function, the coordination number of each ion in the first nearest neighbor shell can also be obtained by integrating the first peak in the distribution curves. The integrated results indicate that each ion is, on average, surrounded by four counterions. The other large peak in the curves at 1.2 nm correlates to the circumferential structure of the confined ion cluster (i.e., peak corresponding to radial distances between cations and anions on the “shell” layer).

In addition to the two large peaks, there are also distinct weak peaks that correspond to longer distances on the profiles at low temperatures, suggesting that the confined ionic liquid does

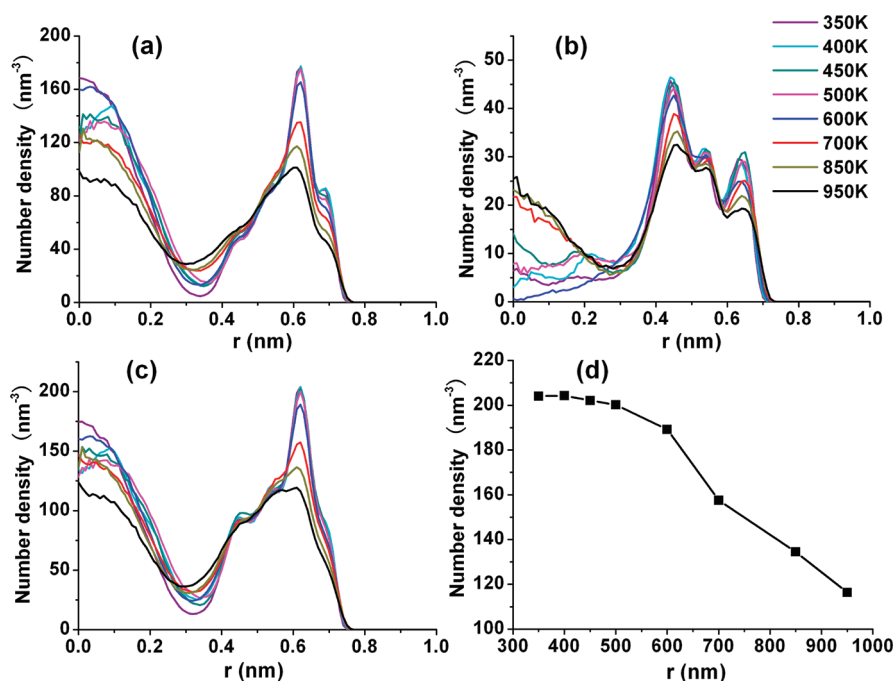


Figure 1. Radial atom number density profiles of (a) cation, (b) anion, and (c) the ionic liquid in the nanopore computed along the pore radius for a series of temperatures averaged over many snapshots. (d) The maxima intensities of the radial number density profiles for the “shell” ionic liquid at different temperatures.

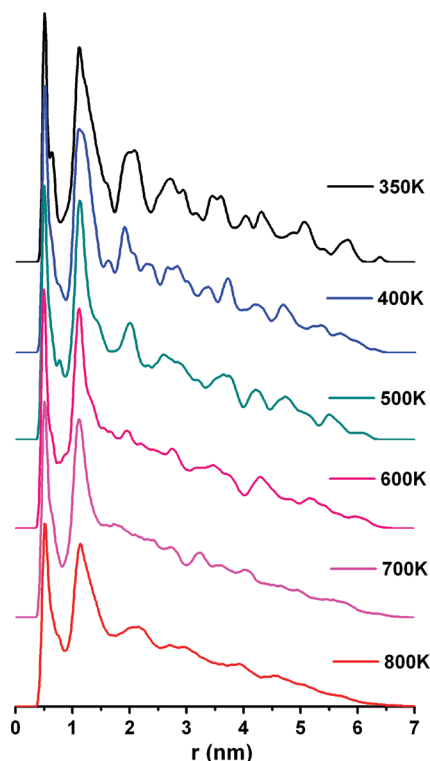


Figure 2. Interionic spacing distribution function of cations and anions confined in a nanotube at different temperatures.

possess some crystalline structural feature. Moreover, this structural feature remains until 500 K, since the constrictions in the molecular movements imposed by confinement have similar effects as the ones due to freezing. As temperature increases

from 500 to 800 K, these local peaks related to the nanocrystal structure gradually disappear, indicating that long-range spatial correlations of the confined ions are completely lost and what we have here are liquids. These significant changes of the nanostructure in the axial direction were also manifested in mass density profiles of the confined ions. The pronounced oscillatory behaviors of mass density profiles in Figure 3a,b are clear signs that ions occupy well-defined locations inside the nanopore and thus form a more crystal-like structure than the liquid below 500 K. For higher temperature in Figure 3c, the oscillatory character of the z -distribution becomes greatly reduced and its magnitude becomes nearly independent of z . This suggests that internal solvation structure lacks crystal-like ordering.

The solid–liquid phase transition of confined liquid can also be detected by monitoring the potential energy profiles.^{15,42–44} It is known that variation in the total potential energy of the system U_{pot} versus temperature T shows different trends at different stages, such as before and during the melting processes and after being molten entirely.⁴⁵ In this work, the potential energy of cations and anions inside the nanopore versus temperature combined with typical snapshots are shown in Figure 4. We observed that as the temperature rises, the potential energy value increases slightly, followed by a dramatic jump. The temperature at this jump merely corresponds to the starting melting temperature T_1 , and another turning point in the $U_{\text{pot}}-T$ curve corresponds to the entirely molten temperature T_2 , observed to be nearly 500 and 800 K, respectively. More impressively, this starting melting point of the nanocrystal is in excellent agreement with the experimentally observed melting point of the [bmim][PF₆] crystals in multiwalled carbon nanotubes.²⁸ The insets in Figure 4 show the radial snapshots of the encapsulated ionic liquids within nanopores before melting and after being molten entirely. The cation and anion distributions in [bmim][PF₆] nanocrystals were separately found to develop

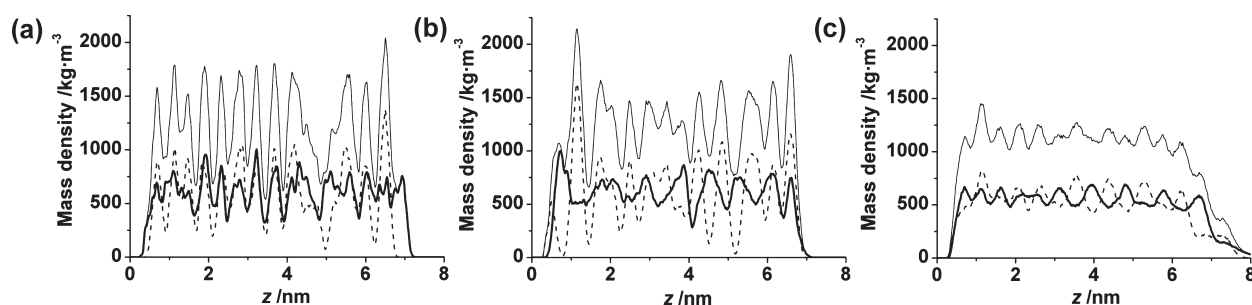


Figure 3. Mass density profiles along the axial direction of the ions confined inside the nanopore at (a) 400 K, (b) 500 K, and (c) 800 K. Gray, black, and dashed lines represent the densities of the ionic liquid, cations, and anions, respectively.

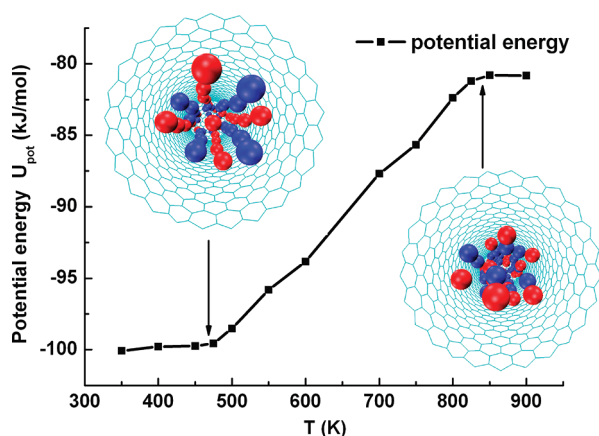


Figure 4. The temperature dependence of confined ions potential energy. The potential energy consists of cation–cation, cation–anion, and anion–anion intermolecular interactions and cation–walls and anion–walls interactions. The insets show the radial snapshots of the encapsulated ionic liquids in nanopores before melting and after being molten entirely. Red and blue balls represent center-of-mass locations of bmim^+ and PF_6^- , respectively.

square structures in a staggered configuration. This structural feature is quite similar to the simulated results of Kim et al.³¹

Ionic Packing of Confined Ion Cluster. To gain insight into the ionic packing during the solid–liquid phase transition, we have analyzed the orientation of the imidazolium ring of cations in the “shell” layer (Figure S2 in the Supporting Information). In Figure 5, we present the probability distribution $P(\theta)$ of angle θ between the imidazolium ring normal vector and the radial vector from the nanotube axis to the midpoint of two nitrogen atoms of the ring. It was observed that the imidazolium rings of bmim^+ in the “shell” layer prefers to form a small tilt angle with the internal surface of the nanopore rather than parallel to it. This special packing arrangement could be due to a combination of factors. The primary factor is a strong π – π stacking interaction between the imidazolium ring and the interior wall.³¹ Another possible factor may be a preferential self-organization of planar molecules: for minimizing the system volume, orientating themselves to form a tilt angle with the internal surface. As the peaks on the profiles indicate, in a frozen state, the imidazolium rings hold a stable angle with the internal surface. However, as temperature increases from 600 to 700 K, $P(\theta)$ suddenly broadens, while the preferential orientation of the rings shifts from $\theta \approx 7.5^\circ$ to $\theta \approx 10.5^\circ$. These changes may be understood in light of the fact that before melting, the available volume is small so that

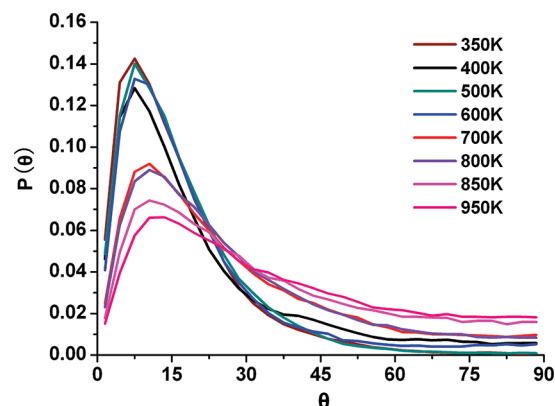


Figure 5. The probability distribution $P(\theta)$ for bmim^+ rings in the “shell” layer of confined ionic liquid; θ is the angle between the ring normal and radial vector from the nanotube axis to the midpoint of two N atoms in the ring.

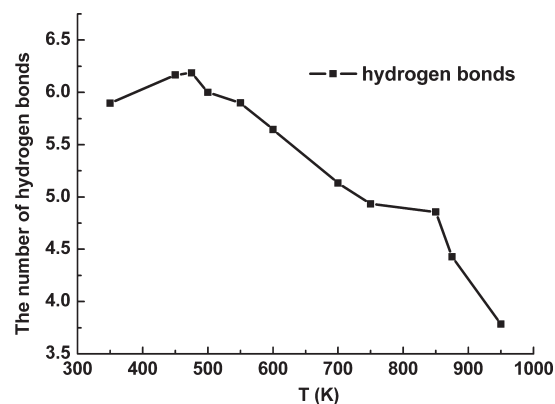


Figure 6. The average number of hydrogen bonds for each anion inside the nanopore for different temperatures.

the bmim^+ rings adopt a relatively ordered configuration. However, at $T > T_1$, the organization of the adsorbate becomes more disordered due to the violent ionic thermal motion and the larger available volume in the nanopore.

Hydrogen bonding is believed to significantly affect the aggregation structure of imidazolium ionic liquids. Generally, if the distance between H and F atoms in cation and anion ($\text{C}–\text{H} \cdots \text{F}$), respectively, is less than $2.67 \text{ \AA}$, then the interaction is recognized as hydrogen bonding.^{46,47} We computed the changes in the number of hydrogen bonds for each confined

anion during the melting transition. In Figure 6, it was found that each anion could form almost six hydrogen bonds with nearby cations in a frozen state. As the amplitude of the ionic vibration increases with temperature, the average number of hydrogen bonds declines linearly. This decline might be directly responsible for the dramatic change in packing arrangement of ions inside the nanopore. These changes in the molecular packing also affect the ionic diffusion behavior (calculation methods and more details are given in the Supporting Information). In the Supporting Information, we mainly analyzed the axial diffusion coefficient D_z of ions during this melting transition. As shown in Figure S3 in the Supporting Information, a dramatic increase in D_z of the ion cluster occurs at the same temperature as the jump in total potential energy and thus marks the melting transition without ambiguity.

CONCLUSIONS

Molecular dynamics simulations of [bmim][PF₆] ionic liquid in nanopores were performed to investigate the structure of confined ion clusters during their melting process. The ordered ionic arrangement in radial and axial directions was observed inside the nanopore, indicating a formation of nanocrystal. The interionic spacing distribution and potential energy profiles revealed that this nanocrystal starts melting around 500 K, which is in excellent agreement with the previous experimentally observed melting point of [bmim][PF₆] crystals in real nanopores.²⁸ Moreover, the diffusion data indicated that in this [bmim][PF₆] nanocrystal, the cations and anions almost freeze around their positions, with each anion possessing almost six hydrogen bonds with nearby cations. As temperature increased beyond 500 K, the average number of hydrogen bonds for each confined anion decline linearly. This decline induced a dramatic change in packing arrangement and finally led to a continuous rise in ion diffusion behavior. This study not only provides many qualitative results on the microstructure of ionic liquid confined in the nanometer scale but also is useful for understanding the thermodynamics of confined ionic liquids.

ASSOCIATED CONTENT

S Supporting Information. Additional tables of data and figures. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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